

2d. Content of Statistics (Optional paper Y542)

Introduction to Statistics.

In **Statistics** learners will explore the theory which underlies the statistics content in A Level Mathematics, as well as extending their tool box of statistical concepts and techniques. This area covers probability involving combinatorics, probability distributions for discrete and continuous random variables, hypothesis tests and confidence intervals for a population mean, chi-squared tests, non-parametric tests, correlation and regression.

5.07 Non-parametric Tests

The concept of a non-parametric test is introduced and explored through the Wilcoxon signed-rank tests and the Wilcoxon rank-sum test (or Mann-Whitney U test), and applied in hypothesis tests concerning population median and identity of populations.

5.08 Correlation

The concept of correlation introduced in A Level Mathematics is formalised and explored further, including the study of rank correlation.